

Dominion Keyboard: Strategic Roadmap to 2030

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Domain: AI Architecture / Mobile Input Methods

Vision: UI interface for the entire Life OS

Executive Summary

The traditional smartphone paradigm—a grid of static apps requiring manual human orchestration—is collapsing. By 2030, mobile computing will be dominated by agentic AI systems that autonomously execute multi-step workflows based on user intent [1] [2]. The keyboard is no longer just a peripheral for entering text; it is the primary conversational and multimodal interface to a user’s personal operating system.

This document outlines the strategic roadmap to transform the Dominion Keyboard from a reactive, cloud-dependent input method into a proactive, agentic super-app. This roadmap is grounded in the philosophy of “living from alignment,” ensuring that the AI architecture supports a fully balanced life across all domains by offloading cognitive burden to autonomous systems.

1. The 2030 Paradigm Shift: Intent-Based Supervisory Control

As identified by UX researchers in 2026, the fundamental shift in human-computer interaction is moving from command-based execution to intent-based outcome specification [3]. Users will no longer type step-by-step instructions; they will state a desired outcome, establish constraints, and delegate execution to the AI.

“The most important thing about AI as an interface is not that it chats in natural language. It is that it changes the user’s role. AI changes computing from command-

based interaction to intent-based outcome specification: the user states the result to be achieved, and the system determines the procedure.” [3]

Dominion Keyboard must evolve to become the orchestration layer for this new paradigm. It will serve as the omnipresent gateway to the user’s digital twin—a comprehensive simulation of the user’s preferences, physiology, and scheduling constraints [4].

2. Architectural To-Do List: The 5-Year Build

To achieve this vision, the Dominion Keyboard architecture must implement four cutting-edge foundations.

2.1 Agent-to-Agent (A2A) Communication Protocol

The keyboard cannot exist in isolation. It must become a node in a federated multi-agent system. By implementing the Model Context Protocol (MCP) and emerging A2A standards, Dominion Keyboard will act as the gateway, translating human intent (via voice or text) into structured JSON payloads that are routed to specialized backend agents (e.g., a scheduling agent, a finance agent, or a smart home agent) [5].

Current State	2030 Target State
Isolated app	Federated gateway node
Direct API calls (OkHttp)	MCP-compliant tool routing
Monolithic logic	Distributed agent swarm

2.2 Proactive Agents

Currently, the keyboard only acts when invoked (reactive). The 2030 architecture requires proactive intelligence. By continuously analyzing the user’s on-screen context via accessibility services and screen reading, the keyboard should initiate actions autonomously. For example, if a user is viewing a flight itinerary, the keyboard should proactively surface a “Book Ride” or “Add to Calendar” intent chip before the user even begins typing [1].

2.3 Digital Twin Foundation and Memory UI

A truly agentic system requires deep, persistent context. The keyboard must interface with a centralized Memory UI Dashboard—powered by Notion as the primary memory store—to build a digital twin of the user. This twin holds the user’s vocabulary, tone, relationships, and constraints. When the keyboard generates a response or executes a task, it queries this digital twin to ensure the action aligns with the user’s overarching life goals (“living from alignment”).

2.4 Multi-Modal Input

The era of typing is dissolving [6]. The keyboard must seamlessly handle voice, text, image, and video input simultaneously. The current implementation requires a manual toggle for the microphone. The 2030 architecture will feature an always-listening, always-watching multimodal input stream that understands context effortlessly.

3. The Path to On-Device Autonomy

The most critical technical hurdle is migrating from cloud dependency to edge computing. Relying on cloud APIs for core input functions introduces unacceptable latency and privacy risks.

3.1 Small Language Models (SLMs) at the Edge

By 2026, sub-billion parameter models (e.g., Gemma 3 270M, SmolLM2) have proven capable of handling practical reasoning tasks entirely on-device [7]. The current Dominion Keyboard relies on an outdated n-gram frequency map for prediction and a cloud API for translation.

Implementation Priority: Integrate a quantized (4-bit) SLM via ExecuTorch or MediaPipe. This model will provide zero-latency, context-aware next-word prediction, inline grammar correction, and tone adjustment without ever sending a keystroke to the cloud.

3.2 Federated Learning for Privacy

To personalize the on-device model without compromising privacy, Dominion Keyboard will implement Federated Learning with Differential Privacy [8]. The local model will learn from the user's specific typing patterns and digital twin data, but only encrypted, aggregated weight updates will ever leave the device.

4. Implementation Phases

To pass the system development quality gate, the current architecture must be solidified before advancing to the agentic phases.

Phase 1: Foundation and Edge Migration (Months 1-6)

- Replace the XML view hierarchy with a custom `SurfaceView` for 60 FPS rendering.
- Implement `setComposingText()` for fluid inline autocorrect.
- Replace the n-gram engine with an on-device quantized SLM (e.g., Llama 3.2 1B or SmolLM2) for prediction and translation.
- Replace cloud Whisper with on-device `whisper.cpp` for zero-latency dictation.

Phase 2: Memory and Digital Twin Integration (Months 6-12)

- Integrate Notion as the primary source for memory pulls.
- Develop the Memory UI Dashboard to allow the user to inspect and edit their digital twin.
- Implement aggressive local data pruning, retaining only aggregated personalization weights.

Phase 3: Agentic Orchestration (Year 2)

- Implement the Model Context Protocol (MCP) within the keyboard service.
- Enable the keyboard to route intents to external agents (e.g., booking, scheduling) rather than just outputting text.
- Develop proactive context awareness via Android Accessibility APIs.

Phase 4: Full Life OS Interface (Years 3-5)

- Finalize the multi-modal input pipeline (vision + voice + text).
 - Achieve full “living from alignment” orchestration, where the keyboard acts as the primary, proactive interface for the entire personal operating system.
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References

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